TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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George Dantzig

- George Bernard Dantzig (1914-2005) was an American mathematician and computer scientist.
- He is best known for his development of the Simplex algorithm, a method for solving linear programming problems.
- Dantzig's work in optimization and operations research has had a profound impact on various fields, including economics, engineering, and logistics.
- He made significant contributions to the theory of linear programming, integer programming, and network flow problems.
- Dantzig's work has been widely recognized and has earned him numerous awards and honors throughout his career.

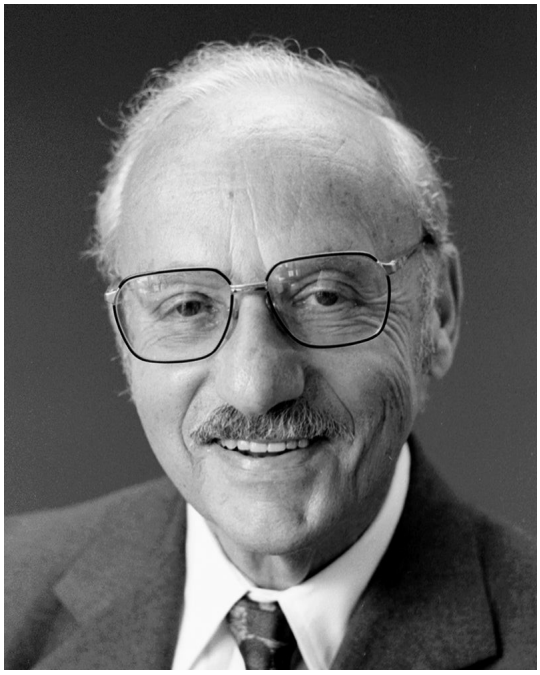


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

The Simplex Algorithm

- It is a "mechanical" method that allows to find optimal solutions to linear programming problems.
- It was invented by George Dantzig in 1947.
- Uses matrix operations until it finds the optimal solution or determines that the problem is unbounded or infeasible.
- It is based in a geometric interpretation of the problem, where the feasible region is a convex polytope and the optimal solution is found at one of its vertices.

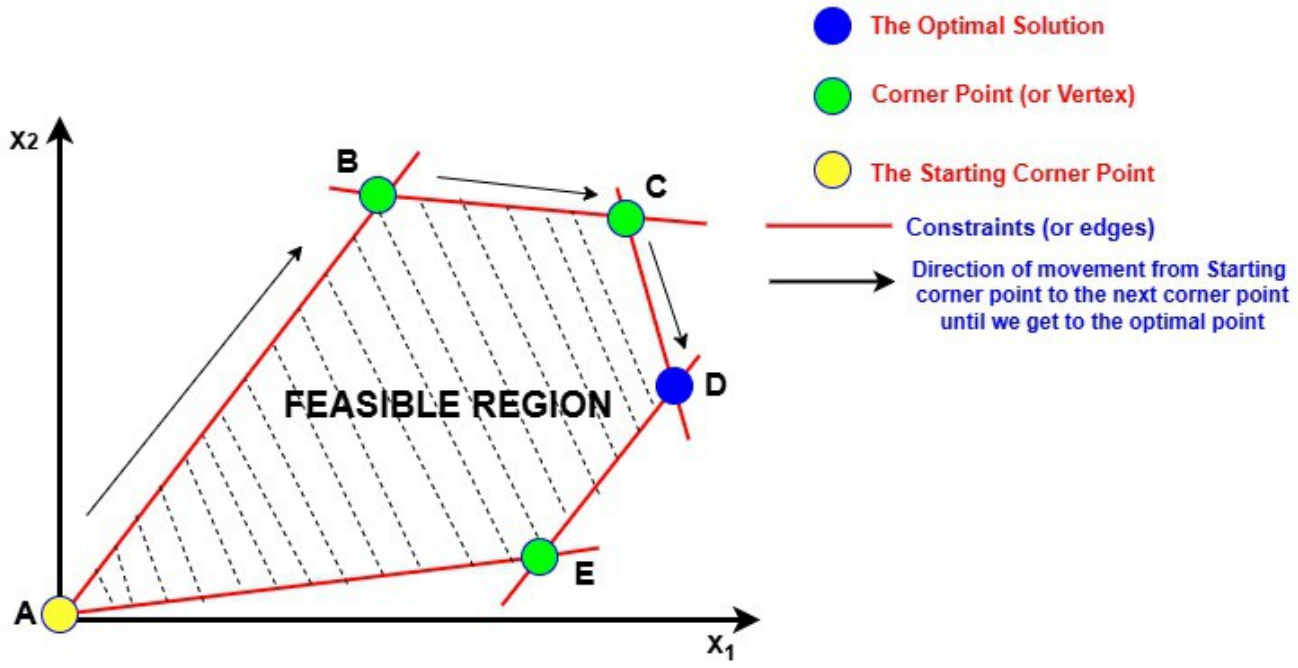


Figure 2: Image inspired by the geometric interpretation of the Simplex Algorithm.

Original Problem

Description

- Problem Name: fdsfdsfds
- Amount of decision variables: 15
- Amount of restrictions: 4
- Total Amount of variables (decision + slack): 19
- Type of problem: Maximization

Equations

$$\text{Max } Z = 6\,x_1 + 3\,x_2 + 6\,x_3 + x_4 + 2\,x_5 + 11\,x_6 + 10\,x_7 + 6\,x_8 + 10\,x_9 + 2\,x_{10} + 5\,x_{11} + 4\,x_{12} + x_{13} + 7\,x_{14} + 4\,x_{15}$$

subject to:

$$4\,x_1 + x_2 + 3\,x_6 + 3\,x_9 \leq 5$$
$$x_2 + x_4 + 2\,x_7 + 3\,x_8 \leq 11$$
$$3\,x_{10} + 2\,x_{11} + 3\,x_{12} + 2\,x_{13} + x_{14} \leq 3$$
$$2\,x_{10} + x_{11} + 2\,x_{12} + 3\,x_{13} + 2\,x_{14} \leq 2$$
$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13}, x_{14}, x_{15} \geq 0$$

Initial Table

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	s_1	s_2	s_3	s_4	b
1.0000	-6.0000	-3.0000	-6.0000	-1.0000	-2.0000	-11.0000	-10.0000	-6.0000	-10.0000	-2.0000	-5.0000	-4.0000	-1.0000	-7.0000	-4.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	4.0000	1.0000	0.0000	0.0000	0.0000	3.0000	0.0000	0.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000	0.0000	5.0000
0.0000	0.0000	1.0000	0.0000	1.0000	0.0000	0.0000	2.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000	11.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	3.0000	2.0000	3.0000	2.0000	1.0000	0.0000	0.0000	0.0000	1.0000	0.0000	3.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	2.0000	1.0000	2.0000	3.0000	2.0000	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000

Final Table

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	s_1	s_2	s_3	s_4	b
1.0000	8.6667	5.6667	-6.0000	4.0000	-2.0000	0.0000	0.0000	9.0000	1.0000	5.0000	-1.5000	3.0000	9.5000	0.0000	-4.0000	3.6667	5.0000	0.0000	3.5000	80.3333
0.0000	1.3333	0.3333	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.3333	0.0000	0.0000	0.0000	1.6667
0.0000	0.0000	0.5000	0.0000	0.5000	0.0000	0.0000	1.0000	1.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000	0.0000	5.5000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	2.0000	1.5000	2.0000	0.5000	0.0000	0.0000	0.0000	0.0000	1.0000	-0.5000	2.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.5000	1.0000	1.5000	1.0000	0.0000	0.0000	0.0000	0.0000	0.5000	1.0000

Solution

Solution

The problem is **unbounded**: there is no finite optimal solution.

Unbounded problem

In iteration 4 the variable x_3 has an improving coefficient in the Z row (-6.0000), but all the entries of its column in the constraint rows are ≤ 0. Therefore the ratio test has no candidate row and x_3 can grow indefinitely without violating any constraint, improving Z without limit. The problem has no finite optimal solution.